

AQGD gradient computations wrong for circuits using one parameter in >

<https://github.com/qiskit-community/qiskit-aqua/issues/588>

ROW 49

AQGD gradient computations wrong for circuits using one parameter in > 1 gate #588

Closed



xmax1 opened on Jul 3, 2019

Informations

- Qiskit Aqua version: 0.5.1
- Python version: 3.7.1
- Operating system: Ubuntu 18

What is the current behavior?

Computes wrong gradient for qaoa.

qiskit > aqua > components > optimizers > aqgd > deriv line 101

Computes the parameter shift rule for the params. Does not include the product rule for when one parameter is used for more than one gate.

Steps to reproduce the problem

Init qaoa circuit

Compute gradient

Compare with actual gradient (using, say, pennylane)

What is the expected behavior?

Uses product rule

Suggested solutions

Use the product rule



chunfuchen added **type: bug** on Jul 3, 2019



chunfuchen on Jul 16, 2019

Contributor

@Nick-Singstock do you have any thought?

woodsp-ibm assigned Cryorls on Jan 17, 2020

Cryorls added **type: feature request** on Mar 20, 2020



woodsp-ibm on Oct 22, 2020

Member

@BryceFuller I believe the new gradient logic that has just been added has product rule support. Presumably this issue could be addressed by allowing AQGD to support a gradient function that was passed in rather than just ignore it as today. Any thoughts - would the gradient need to be typed checked here to ensure it was an analytic gradient? Is this something you might be able to take a look at?

pbark assigned adekusal-drl on Oct 27, 2020



pbark on Oct 27, 2020

Collaborator

Taging also @adekusal-drl to give his opinion on this.

Assignees

Cryorls

adekusal-drl

Labels

type: bug

type: feature request

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants

+1

Zoter

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chunfuchen added **type: bug** on Jul 3, 2019



chunfuchen on Jul 16, 2019

Contributor ...

[@Nick-Singstock](#) do you have any thought?

woodsp-ibm assigned [Cryoris](#) on Jan 17, 2020

Cryoris added **type: feature request** on Mar 20, 2020



woodsp-ibm on Oct 22, 2020

Member ...

[@BryceFuller](#) I believe the new gradient logic that has just been added has product rule support. Presumably this issue could be addressed by allowing AQGD to support a gradient function that was passed in rather than just ignore it as today. Any thoughts - would the gradient need to be typed checked here to ensure it was an analytic gradient? Is this something you might be able to take a look at?

pbark assigned [adekusar-drl](#) on Oct 27, 2020



pbark on Oct 27, 2020

Collaborator ...

Tagging also [@adekusar-drl](#) to give his opinion on this.

woodsp-ibm mentioned this on Jan 27, 2021

[Fix gradient use of the AQGD optimizer Qiskit/qiskit#5650](#)



woodsp-ibm on Jan 27, 2021

Member ...

Closing as there is a issue now opened in Terra to fix this in conjunction with the gradient framework that did not exist at the time.



woodsp-ibm closed this as **completed** on Jan 27, 2021

None yet

Development

No branches or pull requests

Notifications

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Participants

+1

Fix gradient use of the AQGD optimizer #5650

New issue

Closed

#5689



Zoufalc opened on Jan 19, 2021

...

What is the expected behavior?

Although a custom gradient function can be passed to an AQGD optimizer, this custom gradient function is ignored. The optimizer always uses a hard-coded parameter shift implementation which is only compatible with Pauli-Rotations and does neither include the product nor the chain rule.

The AQGD optimizer can be updated such that it uses either the gradient framework to evaluate the gradients or a given gradient function.



Zoufalc added **type: feature request** on Jan 19, 2021



molar-volume mentioned this on Jan 23, 2021

[Fix gradient use of the AQGD optimizer #5689](#)



woodsp-ibm added **mod: algorithms** on Jan 26, 2021



woodsp-ibm on Jan 27, 2021

Member ...

This should take care of the issue as raised a while back in Aqua [/qiskit-community/qiskit-aqua#588](#) (I'll close that issue there off)



Cryoris closed this as **completed** in [#5689](#) on Jan 29, 2021

Assignees

No one assigned

Labels

mod: algorithms

type: feature request

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fix gradient use of the AQGD optimizer
Qiskit/qiskit

Notifications

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